

Finite Elements for Differential Geometry - Exercise 2

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Exercise 5:

Solve Exercise 3.3.6 from the lecture notes on the unit square $\Omega = [0, 1]^2$. Try to incorporate free and simply supported boundary conditions at a boundary part. Does the Ciarlet–Raviart method work in these cases?

Exercise 6:

Solve Exercise 3.5.3 from the lecture notes. To derive the equation for the Airy stress function φ eliminate σ with φ from $\mathcal{D}^{-1}\sigma = \varepsilon(u) + 0.5\nabla w \otimes \nabla w$. Take derivatives $\frac{\partial^2}{\partial x^k \partial y^l}$ of the resulting three equations such that when adding them together u_x and u_y cancel out ($\mathcal{D}^{-1}$ below (3.3.3)).

Exercise 7:

Solve Exercise 3.5.4 from the lecture notes. Find on $\Omega = [0, 1]^2$ $w, \varphi \in H_0^2(\Omega)$ such that

$$\begin{aligned}\Delta^2 w - [\varphi, w] &= f, \\ \Delta^2 \varphi + \frac{1}{2} [w, w] &= g,\end{aligned}$$

where f, g are such that $w = 100x^2(1-x)^2y^2(1-y)^2$, $\varphi = \sin(\pi x)^2 \sin(\pi y)^2$ are the exact solutions.

Hint: Use `CoefficientFunction.Diff(x)` to construct the right-hand sides f and g .

Exercise 8:

Implement hybridization for the TDNNS method for Reissner–Mindlin plates and solve an example on the unit square with one edge clamped, one hard simply supported, one soft simply supported, and one free boundary condition.